

Name: _____ Partner: _____ Period: _____

Part A — Satellite Orbit

A satellite orbits Earth in a circular orbit of radius 8000 km, centered at Earth's center (the origin). It starts at (8000, 0) and travels counterclockwise, completing one full orbit in 2 hours. Using t in hours, the angle after t hours is πt radians.

Tier R — Remediate

A1. Write parametric equations $x(t)$ and $y(t)$ for the satellite's position at time t hours.

$x(t) =$ _____ $y(t) =$ _____

A2. Complete the table. (Round to the nearest whole number where needed.)

t (hours)	$x(t)$ (km)	$y(t)$ (km)	Position (x, y)
0	_____	_____	_____
0.5	_____	_____	_____
1	_____	_____	_____
1.5	_____	_____	_____
2	_____	_____	_____

Tier A — Approaching Proficiency

A3. At what time t (in hours) is the satellite directly above the origin on the north side (i.e., $x = 0$ and $y > 0$)? Show your reasoning. _____

A4. Write new parametric equations for a *clockwise* orbit of the same satellite along the same circular path.

Part B — Rover Path

A rover travels in a straight line from base camp at $(-3, 2)$ to a sample site at $(5, -2)$.

Tier E — Extension

B1. Write parametric equations for the rover's position for $0 \leq t \leq 1$. _____

B2. Find the rover's position at $t = 0.25$ and at $t = 0.75$. _____

B3. Determine the value of t when the rover crosses the x -axis (where $y = 0$). Show your algebra, then state the rover's coordinates at that moment. _____
